Supplementary material for: Two spectral functionals set the exponential cost of shadow moment estimation

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Abstract

This document collects the material the main text refers to but does not print: the full listing of the crossover cells that resolve for the unbiased single-copy estimator, which the main text summarizes but does not tabulate.

S1. THE FULL CROSSOVER TABLE

TABLE I: All 83 crossover cells that resolve for the unbiased U-statistic, the estimator the variance law of the main text describes; the corresponding figure for the range-projected estimator is 72, and the main text reports the two side by side. "mult" is the copy budget as a multiple of the 2000 baseline. A cell counts as within one qubit (✓) when $|\,n^*_{\text{pred}} - n^*_{\text{meas}}\,| \le 1$.

ensemble	k	channel	rate	mult	n^*_{meas}	n^*_{pred}	≤ 1
noisy-pure	2	amp. damp.	0.00	1	2	2	✓
noisy-pure	2	amp. damp.	0.02	1	2	2	✓
noisy-pure	2	amp. damp.	0.05	1	6	6	✓
noisy-pure	2	amp. damp.	0.05	1	6	6	✓
noisy-pure	2	amp. damp.	0.05	4	8	8	✓
noisy-pure	2	amp. damp.	0.10	1	7	7	✓
noisy-pure	2	amp. damp.	0.10	1	7	7	✓
noisy-pure	2	amp. damp.	0.10	4	9	9	✓
noisy-pure	2	dephas.	0.00	1	2	2	✓
noisy-pure	2	dephas.	0.02	1	2	2	✓
noisy-pure	2	dephas.	0.05	1	6	6	✓
noisy-pure	2	dephas.	0.05	1	6	6	✓
noisy-pure	2	dephas.	0.05	4	8	8	✓
noisy-pure	2	dephas.	0.10	1	7	7	✓
noisy-pure	2	dephas.	0.10	1	7	7	✓

ensemble	k	channel	rate	mult	n^*_{meas}	n^*_{pred}	≤ 1
noisy-pure	2	dephas.	0.10	4	9	9	✓
noisy-pure	2	depol.	0.00	1	2	2	✓
noisy-pure	2	depol.	0.02	1	2	2	✓
noisy-pure	2	depol.	0.05	1	2	2	✓
noisy-pure	2	depol.	0.05	1	2	2	✓
noisy-pure	2	depol.	0.05	4	5	5	✓
noisy-pure	2	depol.	0.05	16	7	7	✓
noisy-pure	2	depol.	0.10	1	5	5	✓
noisy-pure	2	depol.	0.10	1	5	5	✓
noisy-pure	2	depol.	0.10	4	7	7	✓
noisy-pure	2	depol.	0.10	16	8	8	✓
noisy-pure	3	amp. damp.	0.00	1	2	2	✓
noisy-pure	3	amp. damp.	0.02	1	2	2	✓
noisy-pure	3	amp. damp.	0.05	1	8	7	✓
noisy-pure	3	amp. damp.	0.05	1	8	7	✓
noisy-pure	3	amp. damp.	0.10	1	8	8	✓
noisy-pure	3	amp. damp.	0.10	1	8	8	✓
noisy-pure	3	dephas.	0.00	1	2	2	✓
noisy-pure	3	dephas.	0.02	1	2	2	✓
noisy-pure	3	dephas.	0.05	1	8	7	✓
noisy-pure	3	dephas.	0.05	1	8	7	✓
noisy-pure	3	dephas.	0.10	1	8	8	✓
noisy-pure	3	dephas.	0.10	1	8	8	✓
noisy-pure	3	depol.	0.00	1	2	2	✓
noisy-pure	3	depol.	0.02	1	2	2	✓
noisy-pure	3	depol.	0.05	1	2	2	✓
noisy-pure	3	depol.	0.05	1	2	2	✓
noisy-pure	3	depol.	0.05	4	4	4	✓
noisy-pure	3	depol.	0.05	16	8	8	✓

ensemble	k	channel	rate	mult	n^*_{meas}	n^*_{pred}	≤ 1
noisy-pure	3	depol.	0.10	1	4	4	✓
noisy-pure	3	depol.	0.10	1	4	4	✓
noisy-pure	3	depol.	0.10	4	8	7	✓
noisy-pure	4	amp. damp.	0.00	1	2	2	✓
noisy-pure	4	amp. damp.	0.02	1	2	2	✓
noisy-pure	4	amp. damp.	0.05	0.25	2	4	×
noisy-pure	4	amp. damp.	0.05	1	8	8	✓
noisy-pure	4	amp. damp.	0.10	0.25	6	6	✓
noisy-pure	4	amp. damp.	0.10	1	8	8	✓
noisy-pure	4	dephas.	0.00	1	2	2	✓
noisy-pure	4	dephas.	0.02	1	2	2	✓
noisy-pure	4	dephas.	0.05	0.25	2	2	✓
noisy-pure	4	dephas.	0.05	1	8	8	✓
noisy-pure	4	dephas.	0.10	0.25	6	6	✓
noisy-pure	4	dephas.	0.10	1	8	8	✓
noisy-pure	4	depol.	0.00	1	2	2	✓
noisy-pure	4	depol.	0.02	1	2	2	✓
noisy-pure	4	depol.	0.05	0.25	2	2	✓
noisy-pure	4	depol.	0.05	1	2	2	✓
noisy-pure	4	depol.	0.05	1	2	2	✓
noisy-pure	4	depol.	0.05	4	3	2	✓
noisy-pure	4	depol.	0.10	0.25	2	2	✓
noisy-pure	4	depol.	0.10	1	3	2	✓
noisy-pure	4	depol.	0.10	1	3	2	✓
haar-pure	2	depol.	0.05	1	2	2	✓
haar-pure	2	depol.	0.10	1	5	5	✓
low-rank	2	amp. damp.	0.05	1	6	6	✓
low-rank	2	amp. damp.	0.10	1	6	6	✓
low-rank	2	dephas.	0.05	1	6	6	✓

ensemble	k	channel	rate	mult	n^*_{meas}	n^*_{pred}	≤ 1
low-rank	2	dephas.	0.10	1	6	6	✓
low-rank	2	depol.	0.05	1	2	2	✓
low-rank	2	depol.	0.10	1	4	4	✓
low-rank	2	depol.	0.30	1	6	6	✓
ghz-noisy	2	amp. damp.	0.05	1	6	6	✓
ghz-noisy	2	dephas.	0.05	1	6	6	✓
ghz-noisy	2	dephas.	0.10	1	6	6	✓
ghz-noisy	2	depol.	0.05	1	2	2	✓
ghz-noisy	2	depol.	0.10	1	5	4	✓
ghz-noisy	2	depol.	0.30	1	6	6	✓